

Heart rate monitoring employing Photoplethysmography

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1. Abstract

In this project, we developed a non-invasive heart rate monitor based on the principle of photoplethysmography (PPG), using optical sensors, filtering techniques, and digital processing to measure heart rate. The design involved integrating a series of stages—optical sensing, signal conditioning, analog-to-digital conversion, and output indication—to create a cost-effective, user-friendly device. We selected specific components, including IR204 and PT204 for sensing, and implemented LM324 active filters and a LM339 comparator for signal processing to ensure accurate heart rate detection. The device was tested across different stages to verify its functionality, demonstrating its ability to accurately convert the physiological signal of heartbeats into a digital output that drives an LED indicator, reflecting real-time heart rate.

Throughout the project realization, our work was supported by an array of essential electronic components and measurement tools. We utilized operational amplifiers, specifically the LM324N, an infrared LED, phototransistor pair, IR204-A and PT204-6B, respectively, to explore various sensor behaviors, and a comparator LM339. We employed the function generator for inputting high voltage signals and AD2 to measure these expected outputs. Our experimental setups were constructed on a breadboard and involved various resistors, potentiometers and capacitors to meet the circuit design requirements. For detailed measurements and analysis, we employed oscilloscopes for waveform observation, multimeters for voltage and current assessments, and the Waveforms software to visualize and analyze the converters behavior effectively.

2. Introduction

In today's technology-driven world, advancements in health monitoring technologies have dramatically transformed how we manage personal health and wellness. These modern services, however, were not always readily available. About 30 years ago, the landscape of medical technology was remarkably different. Back then, sensors and complex electronics were not widely available in healthcare, especially for everyday use. Electrical engineers faced a critical mission: to enhance health outcomes using technology that was not only effective but also accessible and non-invasive for the general public. Their efforts laid the groundwork for many of the convenient solutions we take for granted today.

For our final project in the course ECE 20008, we found ourselves faced with a the same challenge that those pioneers faced more than a decades ago. Our task was to design and construct a heart rate monitor that was non-invasive, cost-effective, and simple enough for anyone to use. The solution we embraced draws upon a well-established technique that dates back nearly three decades—Photoplethysmography

or PPG. This optical method detects changes in blood volume using low-cost, non-invasive optical sensors. By employing this technique, we aimed to create a device that could bring reliable heart rate monitoring to a broad audience without the need for expensive equipment or medical procedures.

In addition to building the heart rate monitor, our project also serves an educational purpose. We aim to understand all the inner workings of health monitoring technologies going through each step of the process. By doing so, we hope to provide insight into the evolution of medical gadgets from luxury items to everyday essentials. This project not only revisits a tried-and-true method but also adapts it for contemporary use, demonstrating the enduring relevance of past innovations in today's high-tech world.

3. Theory

To develop an effective heart rate sensor using photoplethysmography (PPG), it is essential to meticulously design a circuit that is universally applicable and selects components that align with the desired specifications shown in Figure 1. This project divides the heart rate sensor into four critical stages: optical sensing, signal filtering, analog-to-digital conversion, and signal output. Each stage is carefully integrated to process the heart rate signal effectively from start to finish.

Specification	Requirement
Heart rate range	40BPM to 200BPM
Supply Voltage	$\pm 5\text{ V}$
Output Voltage	-5 V to 5 V Digital Signal

Figure 1: Given Specifications for Heart Beat Sensor

For the first stage we employed the IR204 transmitter and PT204 receiver. The IR204 is driven close to its maximum current limit of 100mA to ensure maximum signal strength, as calculated by Equation (1). This equation takes into account the input voltage and the resistor R_D to ensure the diode operates within the current limit and at a safe power specified as 0.25W in the datasheet of the resistors.

$$(1) \quad I_D = \frac{V_{CC} - V_D}{R_D}$$

On the receiver side, the objective is to maximize the DC offset up to the saturation voltage $V_{CE(sat)}$ of 0.4V, as specified, ensuring a better signal detection. The output voltage V_{pt} across the receiver is calculated using Equation (2), which factors in the emitter resistor R_E . This setup allows the receiver to convert the modulated light signals into electrical signals that accurately represent the heart's rhythmic patterns. We can see in Figure 2 the schematic for the entire stage.

$$(2) \quad V_{PT} = i_{PT} R_E$$

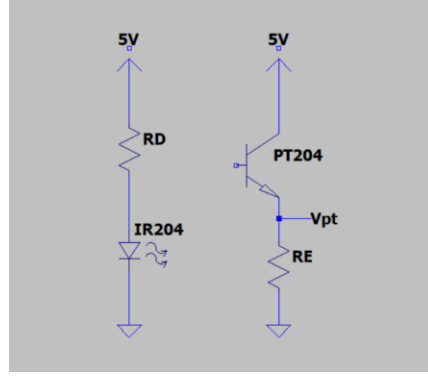


Figure 2: LED driver and Receiver on Stage 1

At this point we have our small ac signal coming from our heartbeat but since it is really small, we need to filter other higher frequency signals that could interfere with our next stages. To refine the sensor signal, two active inverting filters (a high-pass filter and a low-pass filter) are used, both utilizing the LM324 operational amplifier. The filters are designed to pass only the frequencies pertinent to the heartbeat, effectively eliminating unwanted noise. The low-pass filter utilizes Equations (3) and (4), derived from the transfer function presented in Equation (5).

$$(3) \quad k = 1 + \frac{R_2}{R_1} \quad (4) \quad \omega_0 = \frac{1}{R_3 C}$$

$$(5) \quad H(s) = \frac{\left(1 + \frac{R_2}{R_1}\right) \left(\frac{1}{R_3 C}\right)}{s + \frac{1}{R_3 C}}$$

Conversely, the high-pass filter operations are also governed by Equation (3) and (4) since they are in the same inverting configuration, based on the transfer function described in Equation (6). On Figure 3 we can see the circuit schematic we employ for both the high pass and low pass.

$$(6) \quad H(s) = \frac{s \left(1 + \frac{R_2}{R_1}\right)}{s + \frac{1}{R_3 C}}$$

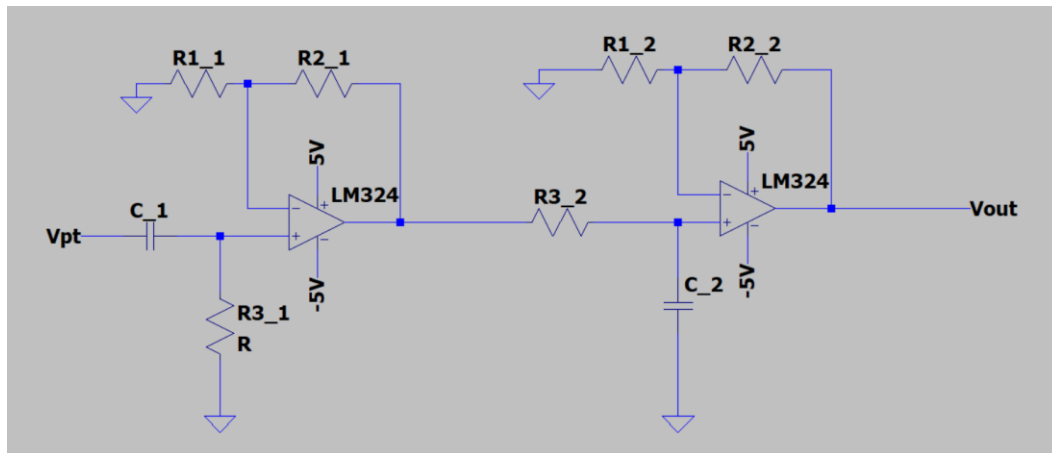


Figure 3: Low Pass and High Pass schematic on Stage 2

To convert the analog to digital signal we employ the LM339 comparator, which simplifies the signal into high and low states based on specific threshold levels. These thresholds are calculated using Equation (7), which establishes the necessary hysteresis, V_{hyst} , linking the resistor values to the input voltages to trigger precise switching at the desired thresholds. On Figure 4 we can observe the followed schematic for this comparator.

$$(7) \quad v_{hyst} = (V_{DD} - V_{SS}) \frac{R_1}{R_1 + R_2}$$

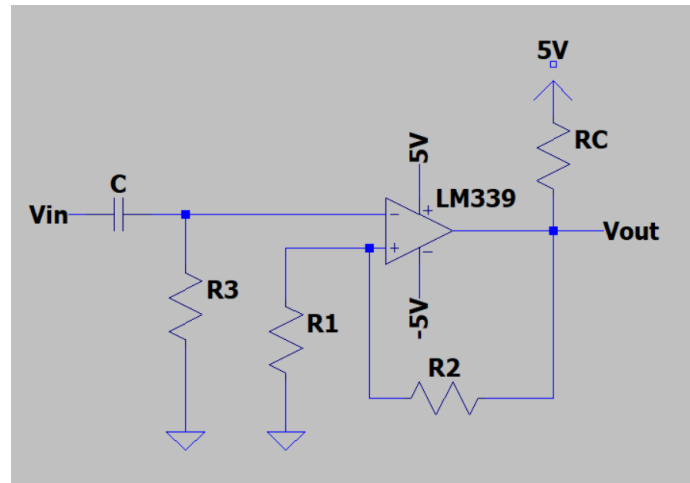


Figure 4: Comparator schematic on Stage 3

The final stage includes an LED indicator, which is activated by the digital outputs from the LM339 to display heart rate pulses visually. To separate the LED control from the main circuit and avoid loading, an N-channel transistor is used as a switch to manage the current flow to the LED. This setup ensures that each pulse is clearly indicated, providing real-time feedback of the heart rate. On Figure 5 we can see the final section of the entire circuit with just the output LED.

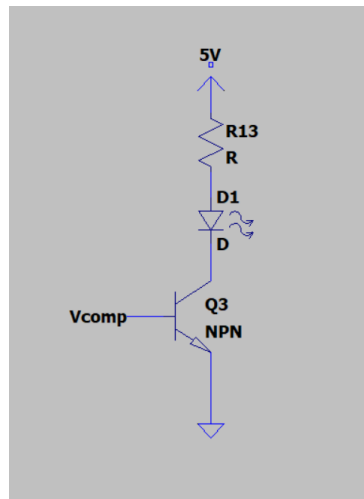


Figure 5: Output LED indicator schematic on Stage 4

4. Design

The initial stage utilized Equation (2) to stay below the saturation voltage $V_{CE(sat)}$ of the phototransistor. Our initial testing indicated that incorporating a 10k ohm potentiometer allowed for fine-tuning the circuit to accommodate various finger thicknesses and skin types, enhancing the circuit's versatility. For the LED driver section, we employed Equation (1) to achieve an optimal current flow through the LED while ensuring it remained under the 100mA maximum rating. We opted for a small resistance value, approximately 45 ohms, achieved by placing two resistors in parallel. This setup yielded a current of 90mA through the LED and a power dissipation of approximately 0.365W in the resistors, maintaining efficiency and safety in the design.

$$I_D = \frac{5-0.95}{45} = 0.09A \quad Power = 0.09 * 4.05 = 0.365W$$

For the next stage the calculations were much more serious. Using Equation (8), we converted the desired heart rate range from beats per minute (BPM) to Hertz, defining our target bandpass filter range from 0.67Hz to 3.33Hz. For the low pass we used Equation (4) with $R_3 = 30k$ ohms and $C = 10\mu F$ to get the desired 0.6Hz. For the high-pass section, we used $R_3 = 47k$ ohms and $C = 100\mu F$. The low-pass filter was designed for a gain of 11, selecting resistors $R_1 = 47k$ ohms and $R_2 = 470k$ ohms. Similarly, the high-pass filter was configured to match this gain. The cumulative gain of these filters reached 121, a product of their individual gains, ensuring substantial amplification of the heart rate signal while attenuating unwanted frequencies.

$$(8) \omega_0 = 2\pi f_0$$

$$k = 1 + \frac{470k}{47k} = 11 \text{ gain} \quad HPF \omega_0 = \frac{1}{47k * 100 \mu} = 0.21 \quad LPF \omega_0 = \frac{1}{30k * 10 \mu} = 3.33$$

For the third stage we focused on processing the filtered signal through an analog-to-digital converter setup. We analyzed the circuit with random values for V_{hyst} and after several trials it gave us 0.9 volts. At the beginning we introduced a variable resistor R_1 as a potentiometer to fine-tune this stage until satisfactory performance was achieved. Equation 7 helped define the limits for our component values, resulting in $R_1 = 1k$, $R_2 = 10k$, and $R_3 = 82k$ ohms. To connect the output to an indicator, an NMOS transistor coupled with a 150-ohm resistor was added to manage the current flow to the LED, ensuring it flashed in sync with the detected heartbeats.

5. Results

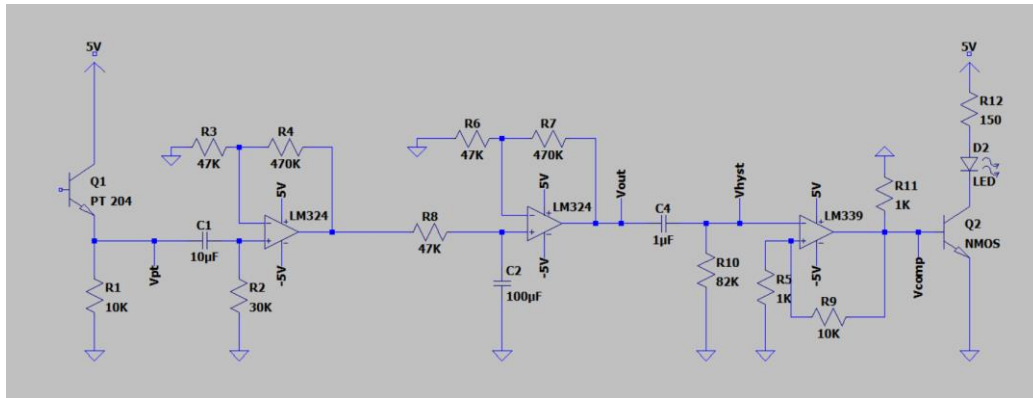


Figure 6: Final Schematic for the heartbeat sensor

After assembling the entire sensor, we conducted general testing to verify that each stage met its design specifications. Let's review the performance of each critical component, starting with the filters. As illustrated in Figures 7 and 8, the filters performed within expected specifications. The high-pass filter demonstrated a cutoff frequency at 0.418 Hz, while the low-pass filter achieved a cutoff at 4.55 Hz, effectively isolating the frequency range pertinent to heart rate signals. Figure 9 shows the output after both filters, highlighting the successful creation of a bandpass profile that accurately captures the heart rate signal's frequency characteristics.

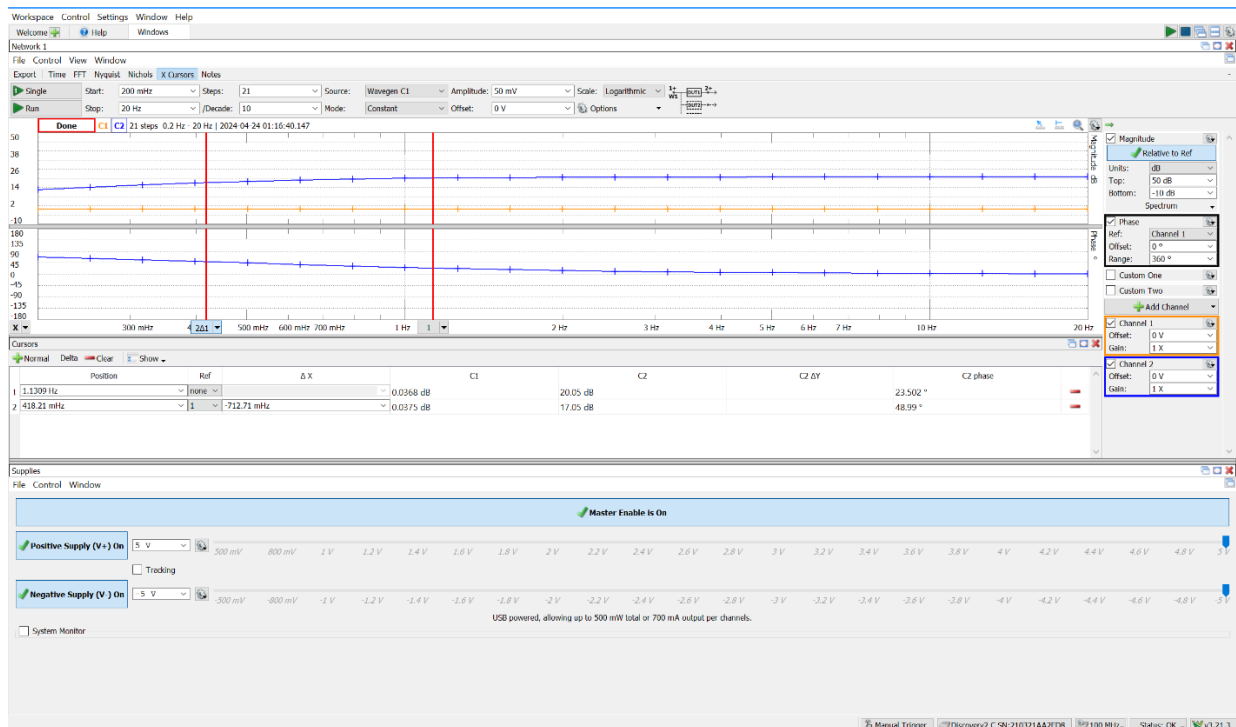


Figure 7: High Pass Filter -3db Cutoff

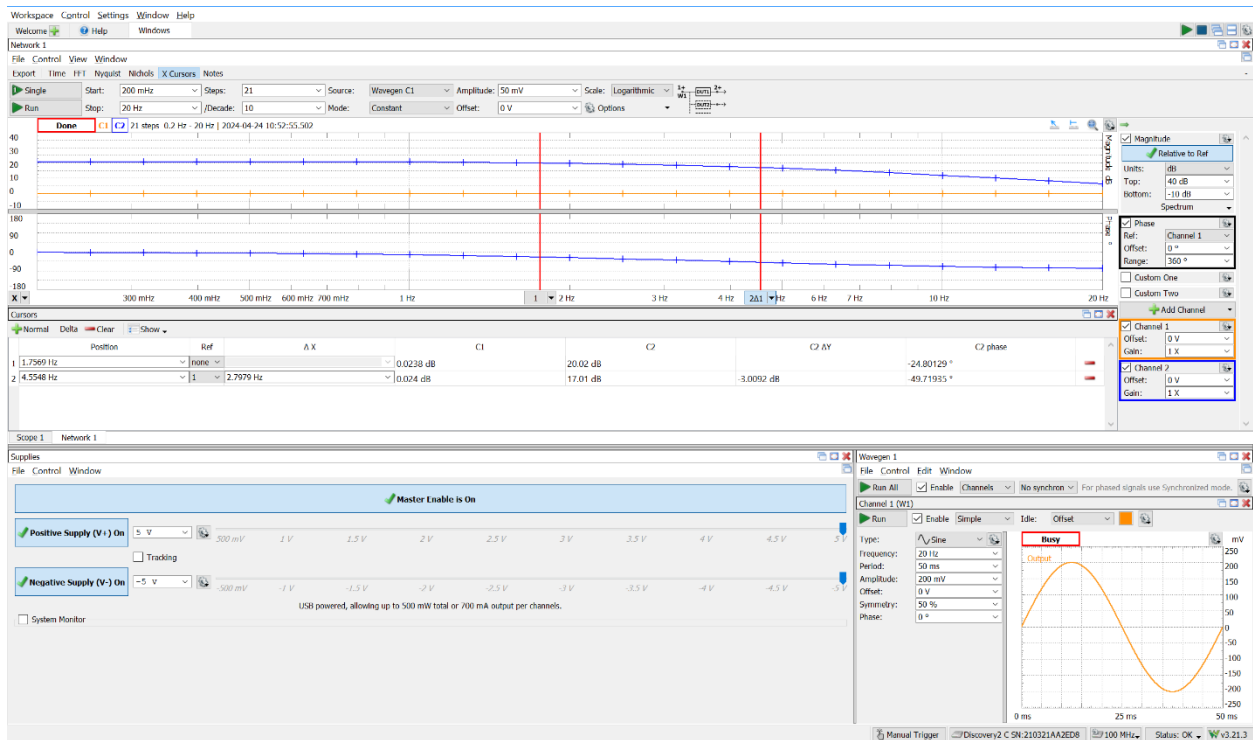


Figure 8: Low Pass Filter -3db Cutoff

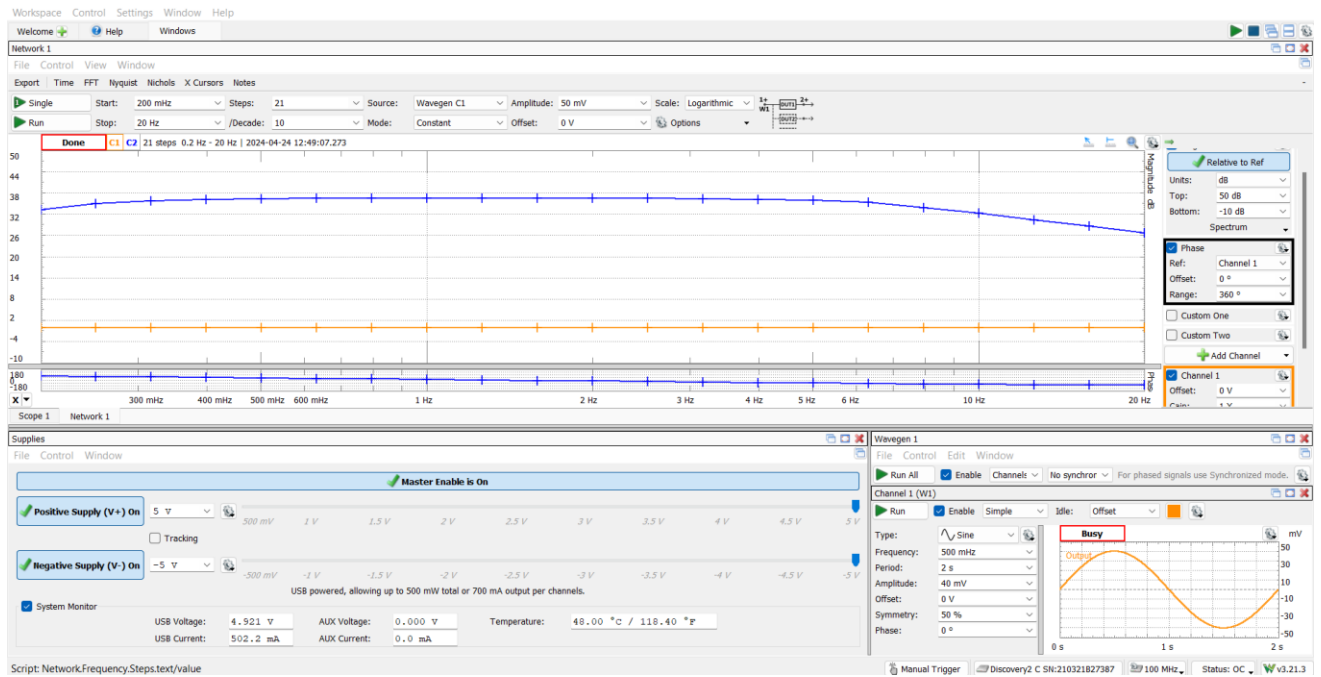


Figure 9: Band Pass Filter (HPF + LPF) -3db Cutoff

Moving on to the analog-to-digital converter, Figure 10 showcases its response to a controlled input. The converter efficiently transformed the filtered analog signal into a clean, periodic digital waveform, matching the frequency and amplitude requirements set for accurate heart rate detection.

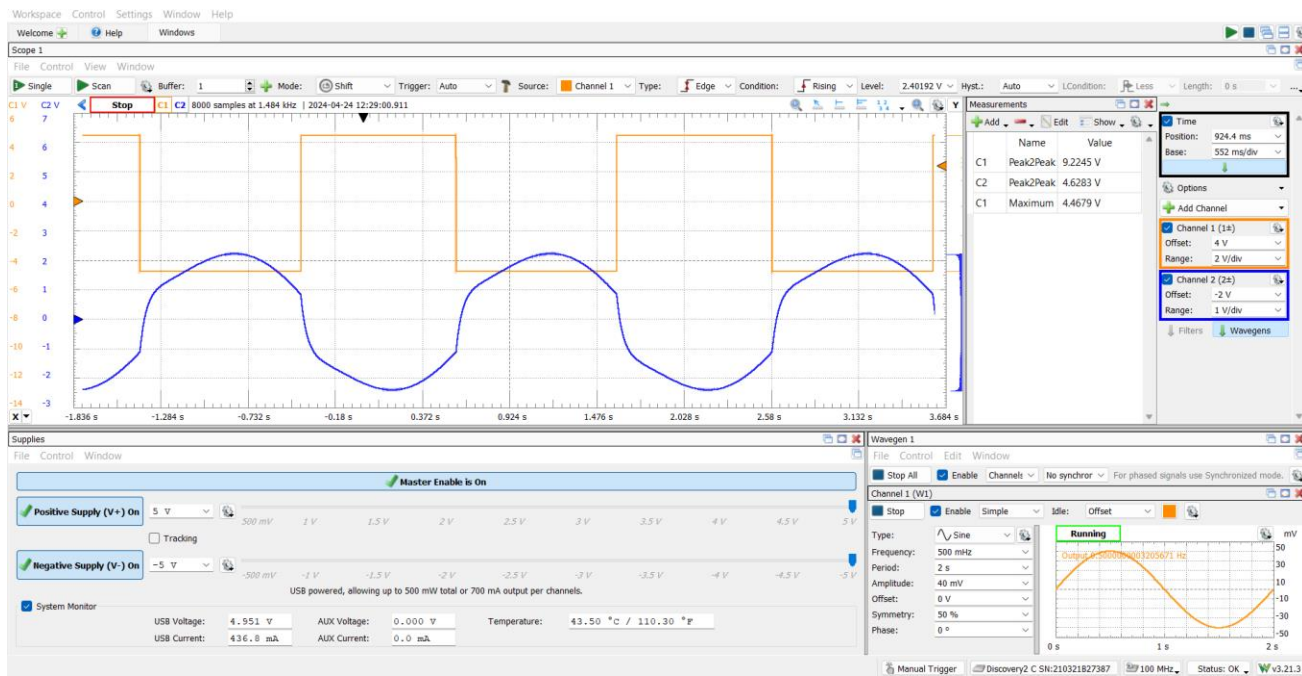


Figure 10: Analog to Digital Converter

The final stage of testing involved assessing the initial sensor output and the overall system integration, displayed in Figure 11. Here, the sensor output before filtering showed a peak-to-peak voltage of 4V, confirming the sensor's sensitivity and appropriateness for capturing physiological signals. After filtering, this signal effectively triggered the analog-to-digital converter, demonstrating the system's capability to process real physiological data and convert it into a digital format suitable for further analysis or display. This comprehensive testing confirms the functionality and effectiveness of each system component, from sensor input through signal processing to digital conversion.

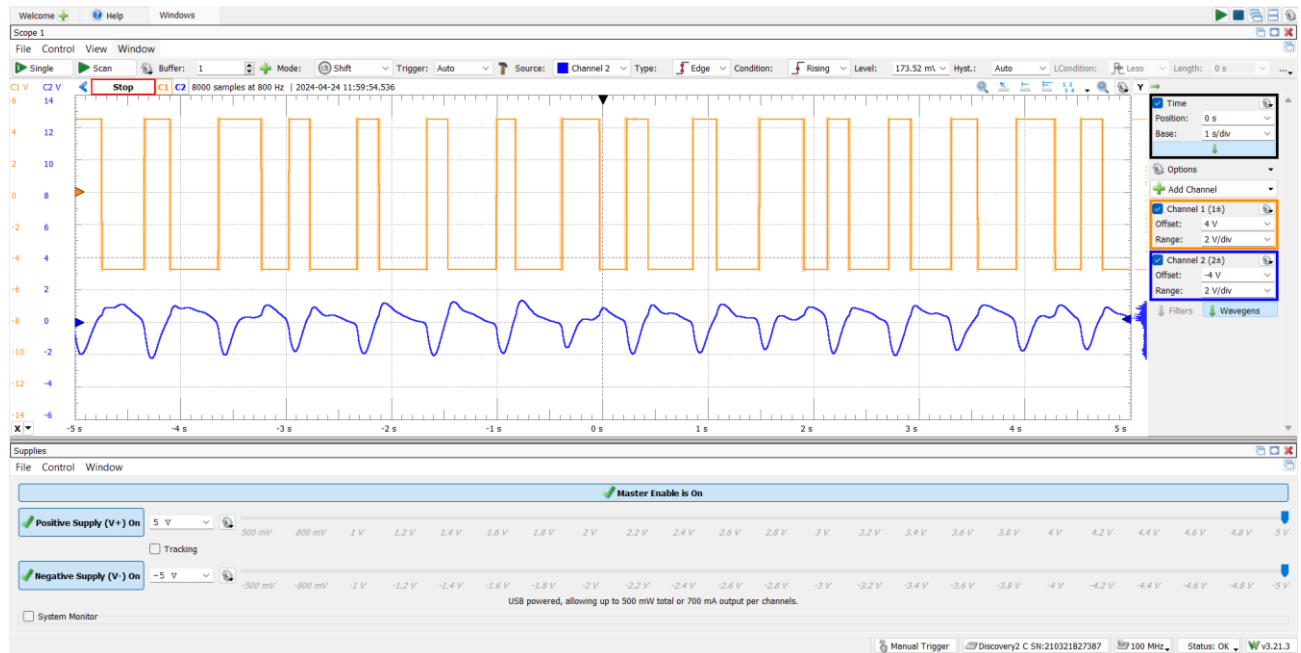


Figure 11: Final Implementation (heartbeat vs comparator output)

6. Conclusion

The completion of this heart rate monitoring project gave us a significant lesson on applying historical technologies to modern applications. By harnessing the principles of photoplethysmography, the project successfully demonstrates how medical monitoring devices can be designed and implemented with affordability and simplicity. The device operates effectively across various stages, from optical sensing through to digital conversion, ensuring precise detection of heart rate. Test results confirmed the functionality of each component, validating the overall design and the integration of the system. This project not only reaffirms the reliability of established methods but also enhances our understanding of designing electronics that are accessible to all kinds of people. Through careful design, execution, and analysis, we have produced a functional device that stands as a testament to the enduring value of combining classical approaches with contemporary engineering education.